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EEE312 Week 14

Implementation of Approximation Method in C Programming Language

Use the least-squares approximation method to approximate the value of P(1.5)from a data set given by (0, 3), (1, 5), (2, -1), (3, 0).
For the problem above, apply the:
(a) linear approximation, and then the,
(b) quadratic approximation.

Beginning of C Code

```
1   // linear least squares approximation
2   #include <stdio.h>
3
4   int main()
5   {
6       float x[4] = {0, 1, 2, 3};
7       float y[4] = {3, 5, -1, 0};
8       float xi = 1.5;
9       float c[2] = {0.0}; // coefficients c0 and c1
10      float A[2][2] = {{0.0}}; // 2x2 matrix A
11      float invA[2][2] = {{0.0}}; // 2x2 inverse matrix A
12      float b[2] = {0.0}; // 2x1 matrix b
13      float P; // polynomial P(x)
```

Line 6-7: Declare and initialize the points (x, y).
Line 8: Declare and initialize the point xi= 1.5. for finding the interpolated point $P(x_i)$.
Line 9: Declare the variables c[0] and c[1] to store the constants of polynomial P(x).
Line 10: Declare a 2×2 matrix A from the matrix equation Ac = b.
Line 11: Declare a 2×2 inverse matrix A from the matrix equation c= $A^{-1}b$.
Line 12: Declare a 2×1 array b from the matrix equation c =$A^{-1}b$.
Line 13: Declare the variable P to store the value of $P(x_i)$.

Linear Approximation

```
15      for(int i=0; i<4; i++)
16      {
17          A[0][0] += 1.0;
18          A[1][0] += x[i];
19          A[1][1] += x[i]*x[i];
20          b[0] += y[i];
21          b[1] += x[i]*y[i];
22      }
23      A[0][1] = A[1][0];
```

Line 15-23: Find the matrix A and b based on the equations:

$$\mathbf{A} = \begin{bmatrix} \sum_{i=0}^{m-1} 1 & \sum_{i=0}^{m-1} x_i \\ \sum_{i=0}^{m-1} x_i & \sum_{i=0}^{m-1} x_i^2 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} \sum_{i=0}^{m-1} y_i \\ \sum_{i=0}^{m-1} y_i x_i \end{bmatrix}$$

```
25      float det = A[0][0]*A[1][1]-A[0][1]*A[1][0];
26      invA[0][0] = A[1][1]/det;
27      invA[0][1] = -A[0][1]/det;
28      invA[1][0] = -A[1][0]/det;
29      invA[1][1] = A[0][0]/det;
30
31      for(int i=0; i<2; i++)
32          for(int j = 0; j<2; j++)
33          {
34              c[i] += invA[i][j]*b[j];
35          }
```

Line 25-29: Find the inverse of matrix A using the equation for a 2x2 matrix:

$$A^{-1} = \frac{1}{det}\begin{bmatrix} A[1][1] & -A[0][1] \\ -A[1][0] & A[0][0] \end{bmatrix}$$

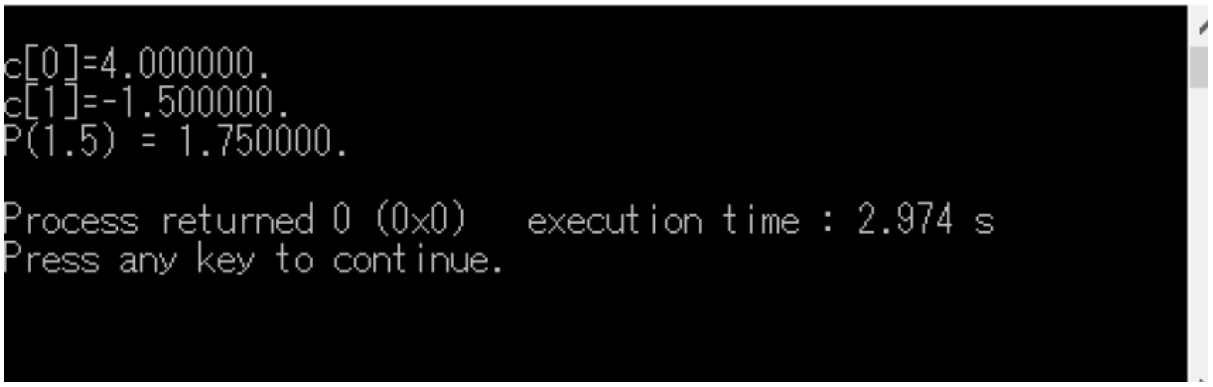
where det= A[0][0]*A[1][1]□A[0][1]*A[1][0].

Line 31-34: Find c[0] and c[1]using the matrix equation:

```
37      P = c[0] + c[1]*xi;
38
39      for(int i=0; i<2; i++)
40          printf("\nc[%d]=%f.", i, c[i]);
41
42      printf("\nP(%.1f) = %f.\n", xi, P);
43
44      return 0;
45  }
```

Line 37: Calculate P(1.5)using the equation:
$P(x) = c_0 + c_1 x$ when x= 1.5
Line 39-42: Print out all resuts at the program output.

Output


c[0]=4.000000.
c[1]=-1.500000.
P(1.5) = 1.750000.
Process returned 0 (0x0)   execution time : 2.974 s
Press any key to continue.